

# Using models to learn about birds and cetaceans

Add your name and student number here

09 July 2026

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This is an example of an autocorrection assignment for Ecological Statistics.

## Instructions

Please read carefully these instructions. **There are some new bold bits you have not seen before.** Failing to do so might lead to you not having the grade you expect.

Never ever under any circumstances:

- change code that is in a chunk that starts with the comment “# do not touch this code chunk”
- change the names of objects created by the template's code
- delete any objects created by the code already on the template
- create objects with the same names as those already present in the template

You should read the assignment on its `.html` version, since that is the only way to be able to see what are some of the dynamic values that your tasks depend on. Currently, many of these dynamic values just take the default value `NA`. When you complete the assignment, if you did all you should as you were expected, there will be no longer any `NAs` shown in the output.

**Very important:** For all numeric results, do not use numbers with longer decimals, as here

```
pi
```

```
## [1] 3.141593
```

You must round all **numerical** values to two decimal places, as here

```
round(pi,2)
```

```
## [1] 3.14
```

Also, **you should make sure that numeric results are indeed numeric**. Therefore, for safety, always consider using the `as.numeric` function for safety, as in

```
as.numeric(round(pi,2))
```

```
## [1] 3.14
```

**WARNING:** Failing to follow any of the above simple instructions might mean you will have a grade of 0 on the assignment.

## Before you begin

Replace the number 11111 in the next chunk with your student number.<sup>1</sup>

Compile the .Rmd after you replaced 11111 by your student number in the code chunk above.

```
#replace 11111 by your student number  
naluno <- 11111
```

From now on, your results are yours alone, and you are ready to start the work itself.

## Introduction

In this assignment you will be mostly working with data from the paper

Shannon G., Layton-Matthews K., Lorentsen S.-H., Mattisson J., Christensen-Dalsgaard S. & Dehnhard N. 2025. Loggers affect the foraging behaviour and fitness of European shags. *Journal of Avian Biology*. DOI: (10.1002/jav.03534)[<https://nsojournals.onlinelibrary.wiley.com/doi/full/10.1002/jav.03534>]

In Shannon *et al.* (2025) the authors explore how animal borne loggers impact European shags, in particular their foraging behaviour and fitness.

---

<sup>1</sup>As you now know if you did the first assignment, the first thing you will do here is to set up a random seed, which will always be unique to you: your student number.



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## Loggers affect the foraging behaviour and fitness of European shags

Graeme Shannon✉, Kate Layton-Matthews, Svein-Håkon Lorentsen, Jenny Mattisson, Signe Christensen-Dalsgaard, Nina Dehnhard

First published: 20 November 2025 | <https://doi.org/10.1002/jav.03534>



SECTIONS

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### Abstract

Biologgers enable real-time collection of detailed behavioural and physiological data from wide-ranging animals, including seabirds inhabiting remote regions. However, the number of studies using tracking devices has not been matched by research exploring the behavioural and fitness costs of deployment, and the effects on data reliability. We assessed how GPS loggers, video loggers, and time depth recorders (TDRs) affect the behaviour, physiology, and reproductive performance of European shags *Gulosus aristotelis* breeding on Sklinna, Norway. The loggers varied in mass, attachment location and deployment duration, allowing comparison of their relative effects. Birds without loggers served as controls to assess logger-related changes in adult body mass, chick growth, reproductive success, and survival. Birds with longer-term tail-mounted GPS loggers and leg-mounted TDRs showed altered foraging behaviour, including shorter trips and dives, compared to birds with tail-mounted GPS and TDRs of the same weight, instrumented for only two days. A mean loss in body mass was experienced by adult birds regardless of the logger type used, while chick growth rate dropped to 20% of that observed in control nests when video, TDR and GPS loggers (~4% of body mass) were deployed together. Logger attachments did not impact reproductive success, and overall logger birds showed higher survival than controls. However, female survival was lower than that of males among birds fitted with video and long-term GPS loggers. Our results demonstrate the importance of measuring behavioural and physiological effects that can scale over time. The advances in our understanding of animal ecology and behaviour generated by biologging have been impressive, but there is a need to consider the impacts on animal welfare and data quality. Consistent reporting of logger deployment details is essential to assess biologging impacts across species and refine protocols that account for device weight, drag and attachment location.

Figures References Related Information

### Metrics

Am score 2

### Details

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### Keywords

biologging conservation  
monitoring movement ecology  
seabird tag effects

### Publication History

Issue Online:  
20 November 2025

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20 November 2025

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24 September 2025

The pdf of the paper is on moodle under this assignment folder, so that you can look at additional details if you want. Here you will be only analysis one of the several datasets the authors provide, namely `Foraging_trip.csv`.

These are the columns descriptions, taken from the authors readme file available at the dryad repository with DOI: [10.5061/dryad.2z34tmq07](https://doi.org/10.5061/dryad.2z34tmq07). It is very good for science that scientists provide their data so that others can do additional science with those, and checking the results are correct. This is one of the ways we can make science more transparent and reproducible.

### 1. Foraging behaviour (dataset: `Foraging_trip.csv`)

| Variable | Definition                                |
|----------|-------------------------------------------|
| trip_id  | Unique ID for each recorded foraging trip |
| sex      | Male or Female                            |

| Variable | Definition                                     |
|----------|------------------------------------------------|
| year     | Year                                           |
| TripComp | Trip completed                                 |
| N        | Number of GPS locations                        |
| maxdist  | Maximum distance from the colony in metres     |
| start    | Start date and time (format: DD.MM.YYYY HH:MM) |
| end      | End date and time (format: DD.MM.YYYY HH:MM)   |
| length   | Length of trip in decimal hours                |
| tag      | Type of logger used (IgotU, pathT or video)    |
| duration | Time since the logger was deployed in days     |

Each record corresponds to a foraging trip.

## Reading the data

Read the data in the next code chunk, naming the object that will hold the data `shagsALL`.

```
#this code should be a single line of code
# this is the only code chunk you edit in this section
shagsALL <- NA
# once you have read the data with your own code in the line above,
# delete the line of code just below here, which is there just so that this
# document compiles before you do assignment
shagsALL <- data.frame(N=1:10000,shags=NA)
```

## Don't touch this (section)

Nothing in this section is to be touched. Therefore, just... don't.

You are not supposed to understand what this code is doing, so if you do not, that's absolutely fine.

The code is simply generating datasets that are unique to you, which you will then work on in the next sections.

Here we pretend the hard work the authors had in the field was jeopardized by a lack of backups. A file became corrupted and some of the data was lost. Never forget, do not be these people: always have data backups!

```
set.seed(naluno)
n <- round(runif(1,1000,nrow(shagsALL)))
shagsR <- shagsALL[sample(1:nrow(shagsALL),size = n,replace=FALSE),]
```

The file with the records that were recovered is `shagsR`. This is the data you will work on, unless you are explicitly told otherwise.

The original dataset had 10000 records. Your dataset is now a random subset of those observations, stored in an object named `shagsR`, of class `data.frame`. If this worked fine it should be a `data.frame`.

## Your shags analysis

### Optional

This subsection is optional, but since you learn by doing, I recommend you do it!

Any statistical tests or models should only be implemented after a thorough exploration of the data.

Feel free to explore the data in `shagsR`. If you produce any interesting insights from an exploratory data analysis - i.e. without using any tests, just data visualizations - describe them below. These can be used later if you want to convince me you worked extra hard ;)

```
# makes some plots about the data  
# draw some conclusions from them
```

## Mandatory

How many records were not destroyed?

```
# replace the value NA with an expression that  
# returns the desired value  
result1 <- NA
```

A total of NA records were not destroyed.

What was the proportion of the data that was lost, in terms of the number of records?

```
# replace the value NA with an expression that  
# returns the desired value  
result2 <- NA
```

The proportion of the data that was lost was NA.

Represent in a figure the number of records per tag type. What was the tag type with the largest number of records?

```
# replace the value NA with an expression that  
# returns the desired value  
result3 <- NA
```

The tag type with the largest number of records was NA.

Evaluate if the distribution of the tag types is the same across years. What is the value of the test statistic of the adequate statistical test to do so?

```
result4 <- NA
```

The value of the test statistic is NA.

What is the decision if you consider the significance level of 0.001

```
result5 <- NA  
# uncomment the line that is correct for your dataset  
#result5 <- "reject H0"  
#result5 <- "not reject H0"  
#result5 <- "accept H0"  
#result5 <- "accept H1"
```

At the significance level of 0.001 we should NA.

How might this result influence the results of comparisons across different tag types?

Is there a relation between the maximum distance and the length of the trip? Make a suitable plot that looks at this relation, add a minimum square line to the plot, and calculate the correlation between the two variables, which you should add to the plot

```
# add the correlation between the two variables here  
result6 <- NA
```

The linear correlation between the two variables is NA.

Do you reject or do you not reject the null hypothesis that the correlation is 0 at the level of significance 0.07?

```

# add your decision here
# comment this line and then
result7 <- NA
# uncomment the line that is correct for your dataset
#result7 <- "reject H0"
#result7 <- "not reject H0"
#result7 <- "accept H0"
#result7 <- "accept H1"

```

At the significance level of 0.07 we should NA.

How do you interpret this relation?

Model the length of the trip as a function of the tag, sex and year, using a GLM with one of the following distributions for the family: (1) Gaussian, (2) Poisson, (3) beta, (4) Gamma or (5) binomial. Irrespective of the family chosen, consider a log-link function. Also, despite that being a dubious procedure, consider the year as a continuous variable. Call the fitted model `model.length`.

Ignoring interactions, does sex influence the response, at the significance level 0.001?

```

# add your decision here
# comment this line and then
result8 <- NA
# uncomment the line that is correct for your dataset
#result8 <- "reject H0"
#result8 <- "not reject H0"
#result8 <- "accept H0"
#result8 <- "accept H1"

```

At the significance level of 0.001, for sex, we should NA.

What about the year?

```

# add your decision here
# comment this line and then
result9 <- NA
# uncomment the line that is correct for your dataset
#result9 <- "reject H0"
#result9 <- "not reject H0"
#result9 <- "accept H0"
#result9 <- "accept H1"

```

At the significance level of 0.001, for year, we should NA.

Given your model, what will be the predicted value for the length of a trip for a female, in 2021, with a tag of the type `pathT`.

```
result10 <- NA
```

The predicted value is NA.

Why is it that year might not be well represented by a continuous variable? In such a case, what should it be considered?

```

# add your decision here
# comment this line and then
result11 <- NA
# uncomment the line that is correct for your dataset
# result11 <- "logical"
# result11 <- "numeric"

```

```
# result11 <- "factor"  
# result11 <- "binary"
```

Year should be considered as a NA variable. Why is that?

## Passive acoustic monitoring

The dataset in file `MMSL2R_FINAL_202505.csv` was used in Taylor *et al.* (2025) to assess the yearly occurrence of cetaceans at a single location in Australia. Each row in the dataset corresponds to a single record and (here we assume) all records have the same duration.

*Marine Mammal Science*



### ARTICLE

## Passive Acoustic Monitoring Reveals Temporal and Environmental Drivers of the Cetacean Community in the Southern Great Barrier Reef

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### ABSTRACT

Monitoring cetaceans in Australia's Great Barrier Reef (GBR) is essential for understanding their ecological roles and informing conservation, yet traditional visual surveys are limited by logistical constraints. Passive acoustic monitoring (PAM) offers a complementary solution, particularly in acoustically complex coral-reef habitats of the southern GBR. Over 2 years, we deployed an autonomous recorder at Lady Elliot Island to document cetacean presence and assess environmental associations. We detected vocalizations from humpback whales (*Megaptera novaeangliae*), dwarf minke whales (*Balaenoptera acutorostrata* subsp.), and delphinids, likely spinner (*Stenella longirostris*) and/or bottlenose dolphins (*Tursiops* spp.). Humpback whale song dominated recordings from June through October, while dwarf minke whale "Star-Wars" calls occurred from May to August, marking the first acoustic evidence of this subspecies within the southern GBR. Dwarf minke whale detections correlated positively with chlorophyll-*a* concentration, whereas humpback whale detections were negatively associated with wind speed. Rorqual vocalizations declined during periods of elevated anthropogenic noise, while odontocete detections remained unaffected. These results highlight PAM's effectiveness for detecting cetaceans within complex reef soundscapes and suggest acoustic niche partitioning among taxa. These critical baseline data inform management strategies for understudied GBR cetacean populations and establish a robust framework for future long-term monitoring efforts and effective conservation actions.

Read the data in the next code chunk, naming the object that will hold the data `taylor`.

```

#this code should be a single line of code
# this is the only code chunk you edit in this section
taylor <- NA
# once you have read the data with your own code in the line above,
# delete the line of code just below here, which is there just so that this
# document compiles before you do assignment
taylor <- data.frame(N=1:10000,pam=NA)

```

## Don't touch this (section) either

Nothing in this section is to be touched. Therefore, just... don't. You are not necessarily supposed to understand what this code is doing, so if you do not, that is absolutely fine. The code is simply generating datasets that are unique to you, which you will then work on in the next sections.

Here we again pretend the hard work the authors had in the field was jeopardized by a lack of backups. The file became corrupted and some of the data was lost. This cautionary tale can't be stressed enough: always have data backups!

```

set.seed(naluno)
corrupted <- sample(1:nrow(taylor),size = 100,replace=FALSE)
taylorR <- taylor[-corrupted,]
sps <- c("Dolphin","Minke","Humpback")
mysp <- sample(sps,1)
# a couple of random days of the year
julian_days <- sample(1:365,2,replace = FALSE) # example Julian day
date1 <- as.Date(julian_days[1], origin = "2025-01-01")
mymon1 <- month(date1)
monname1 <- month.name[mymon1]
myday1 <- day(date1)
date2 <- as.Date(julian_days[2], origin = "2025-01-01")
mymon2 <- month(date2)
monname2 <- month.name[mymon2]
myday2 <- day(date2)

```

## Your PAM analysis

What is the most frequent species to occur at this site?

```

# add your decision here
# comment this line and then
result12 <- NA

```

The most frequent species to occur is NA.

Now fit a model to the data to predict the presence of Minke as a function of the day of the year (DOY). Call it mymod. Select the most appropriate model components from the options below:

```

# model
# 1. lm
# 2. glm
# 3. gam
# family
# 1. gaussian
# 2. beta
# 3. gamma
# 4. poisson

```



```
# 5. binomial
# 6. negative binomial
# link function
# 1. identity
# 2. log
# 3. logit
# 4. cloglog
# 5. inverse
```

tip: if you obtain warning messages regarding convergence problems, in particular if you are using a model that allows for non-linearity, try to restrain the wiggleness of the model by setting the maximum number of degrees of freedom in your smooth. Something like `s(X,k=6)` should work.

Predict the probability of presence of Minke, for a non-leap year, on February 16:

```
# probability of presence goes here
result13 <- NA
```

The predicted probability is NA. And the same prediction on February 23:

```
# probability of presence goes here
result14 <- NA
```

The predicted probability is NA.

For these two specific days, what is the Julian day for which that probability is the highest?

```
# replace the NA with the correct Julian day
result15 <- NA

# DO NOT TOUCH THIS CODE CHUNK
date.maxp <- as.Date(result15-1, origin = "2025-01-01")
mymon.maxp <- month(date.maxp)
monname.maxp <- month.name[mymon.maxp]
myday.maxp <- day(date.maxp)
```

That happens to be for Julian day NA, or in other words NA NA.

## Very important: do not touch the code below

Do not touch the code below, as it produces a file with all your answers, that you will submit to moodle. Also, do NOT touch the .txt file that gets produced when you compile the document.

You are required to submit 3 files:

- An .Rmd file named `HandInT3_N11111.Rmd`<sup>2</sup>
- The two files created when you compile the .Rmd file
  - the .docx file named `HandInT3_N11111.docx` (note this means you have to compile the .Rmd, i.e. to knit it, to word!)
  - the .txt file `ResT3_N11111.txt` (This .txtfile has a format that you do not need to understand, please do not touch this file; failing to follow this instruction could mean you have a grade of 0)

where in the 3 file names the 11111 will be your student number<sup>3</sup>.

<sup>2</sup>For reasons beyond my imagination in some computers this file might appear under Windows explorer as a file of type R HTML file. It is NOT an HTML file, but an .Rmd file.

<sup>3</sup>Your student number is added by default to the .txt file, but you need to add it manually to the .Rmd and .html files by renaming the files.

## References

- Shannon, G., Layton-Matthews, K., Lorentsen, S., Mattisson, J., Christensen-Dalsgaard, S. & Dehnhard, N. (2025). [Loggers affect the foraging behaviour and fitness of european shags](#). *Journal of Avian Biology*, **2025**.
- Taylor, I.A.R., Beutel, S., Dudgeon, C.L., Armstrong, A.O., Townsend, K.A., Edwards-Francis, R. & Potvin, D.A. (2025). [Passive acoustic monitoring reveals temporal and environmental drivers of the cetacean community in the southern great barrier reef](#). *Marine Mammal Science*, **42**.